

Comprehensive CNN Report for Food Delivery Time Prediction

This report presents the implementation and evaluation of a Convolutional Neural Network (CNN) model for predicting whether a food delivery would be classified as Fast or Delayed. The project involved converting structured tabular delivery data into synthetic spatial image representations for CNN-based learning.

1. Methodology

The dataset was cleaned and preprocessed by handling missing values, encoding categorical variables, and normalizing numerical features. Feature engineering included: Haversine distance calculation Rush-hour feature generation Weather severity analysis The delivery data was transformed into synthetic route images using OpenCV. The generated images included: Customer and restaurant markers Traffic-based route coloring Weather-based backgrounds Route thickness based on delivery distance Spatial heat regions

2. CNN Architecture

The CNN architecture used multiple convolutional and pooling layers followed by GlobalAveragePooling2D and dense layers. BatchNormalization and Dropout were applied to improve stability and reduce overfitting.

3. Validation and Hyperparameter Tuning

5-fold cross-validation was used to evaluate generalization performance. RandomizedSearchCV was used for hyperparameter tuning.

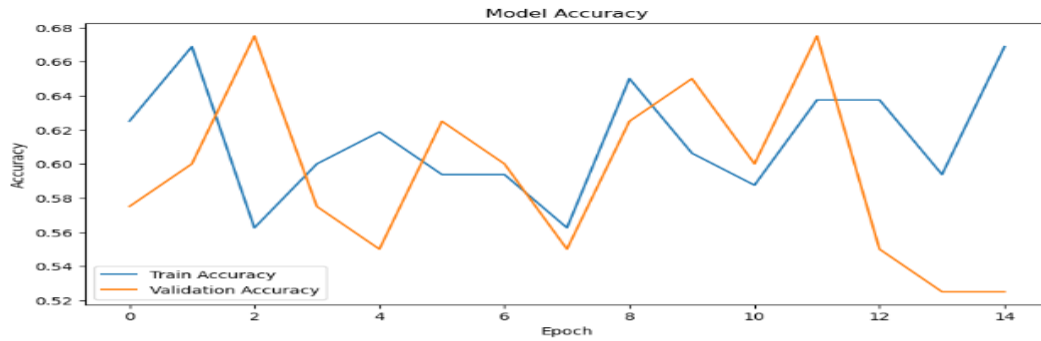
Best Parameters: Filters = 32 Kernel Size = 3 Learning Rate = 0.0001 Epochs = 15 Batch Size = 16 Best Cross-Validation Accuracy: 56.88%

4. Model Performance

Model	Accuracy	Precision	Recall	F1 Score
CNN	52.5%	52.9%	45%	0.486
Logistic Regression	62.5%	64.7%	55%	0.595

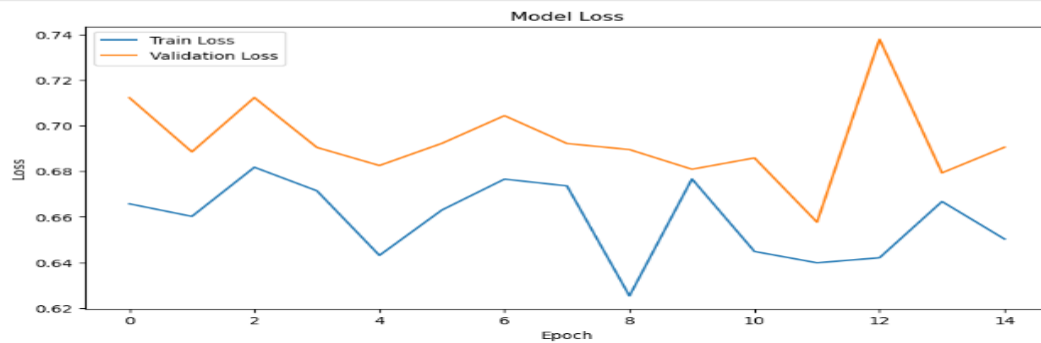
5. Learning Curve Analysis

Learning curves indicated relatively stable training behavior without severe overfitting. However, fluctuations in validation accuracy and loss suggested limited generalization capability due to the small dataset size and simplified synthetic images.



```
3]: plt.figure(figsize=(10,5))
plt.plot(history.history['loss'])
plt.plot(history.history['val_loss'])
plt.title('Model Loss')
plt.ylabel('Loss')
plt.xlabel('Epoch')

plt.legend([
    'Train Loss',
    'Validation Loss'
])
plt.show()
```



6. Key Findings

Enhanced spatial images improved CNN learning capability. Data augmentation improved robustness. Cross-validation revealed moderate generalization limitations. Logistic Regression remained more effective for structured delivery data.

7. Conclusion

The project demonstrated the full workflow of CNN-based delivery prediction using synthetic spatial image representations. Although CNN performance improved substantially after architecture optimization and feature enhancement, Logistic Regression still achieved stronger overall performance on the structured dataset. The work highlighted the importance of selecting machine learning models that align with the underlying data structure.